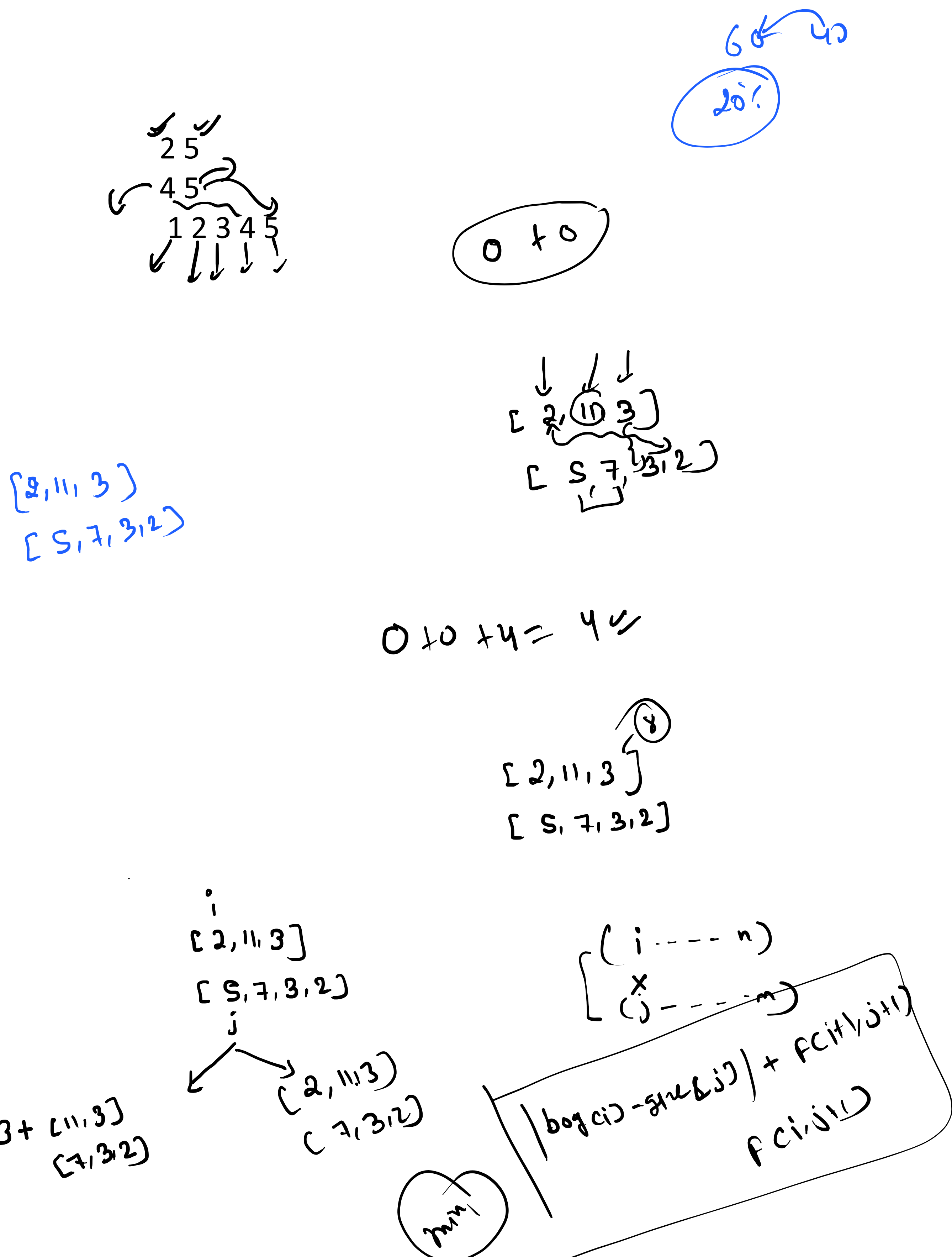


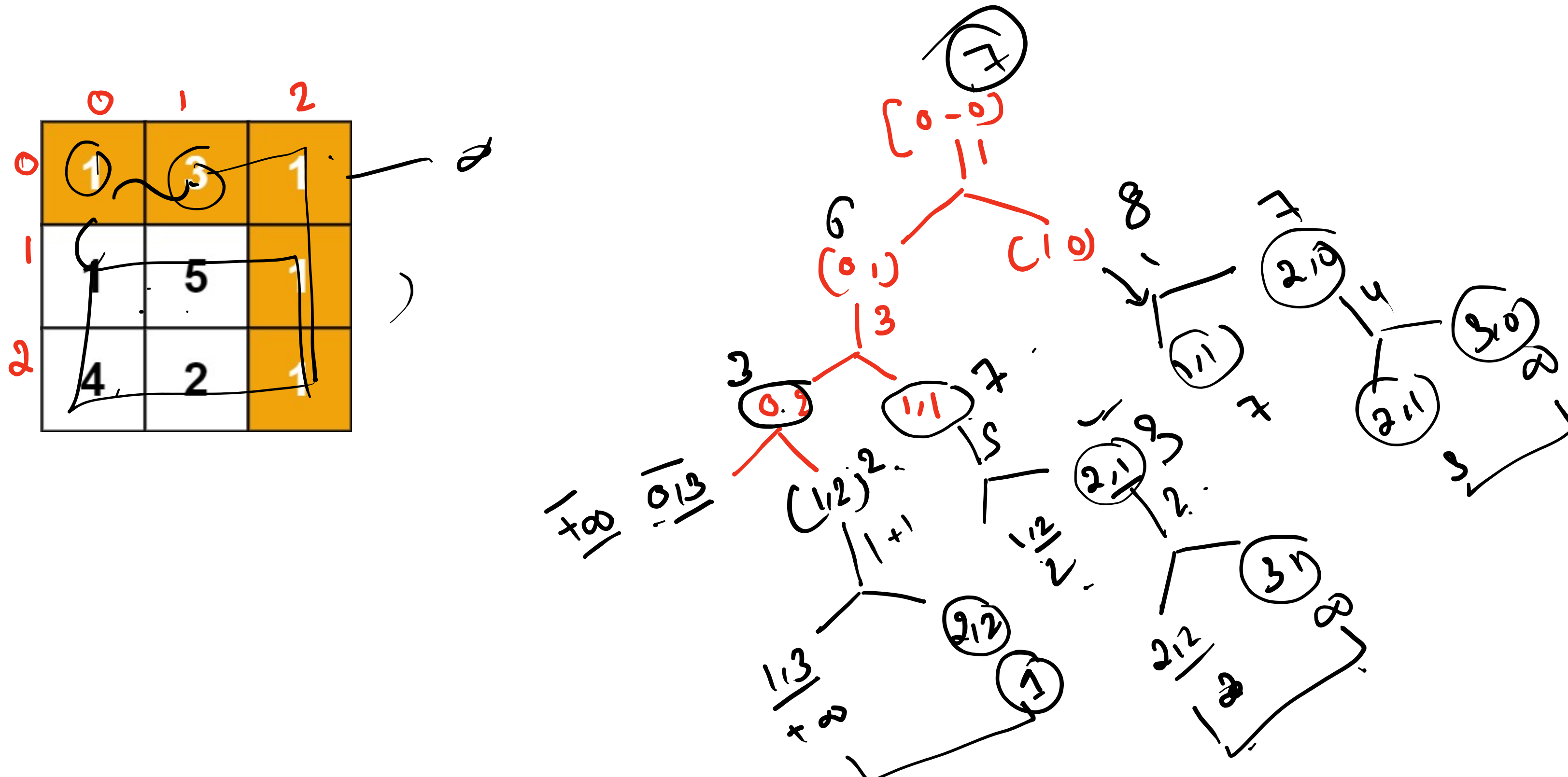
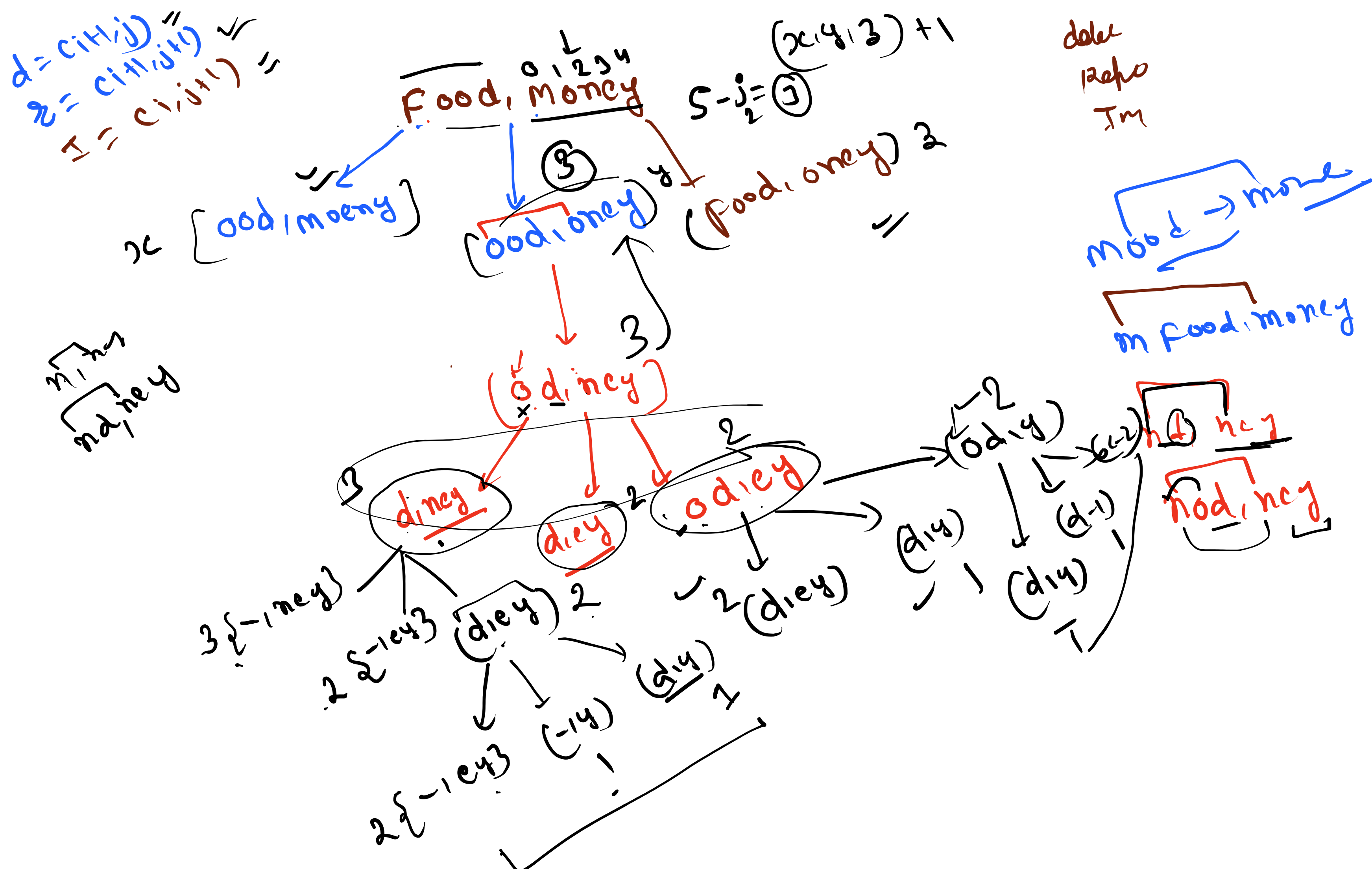
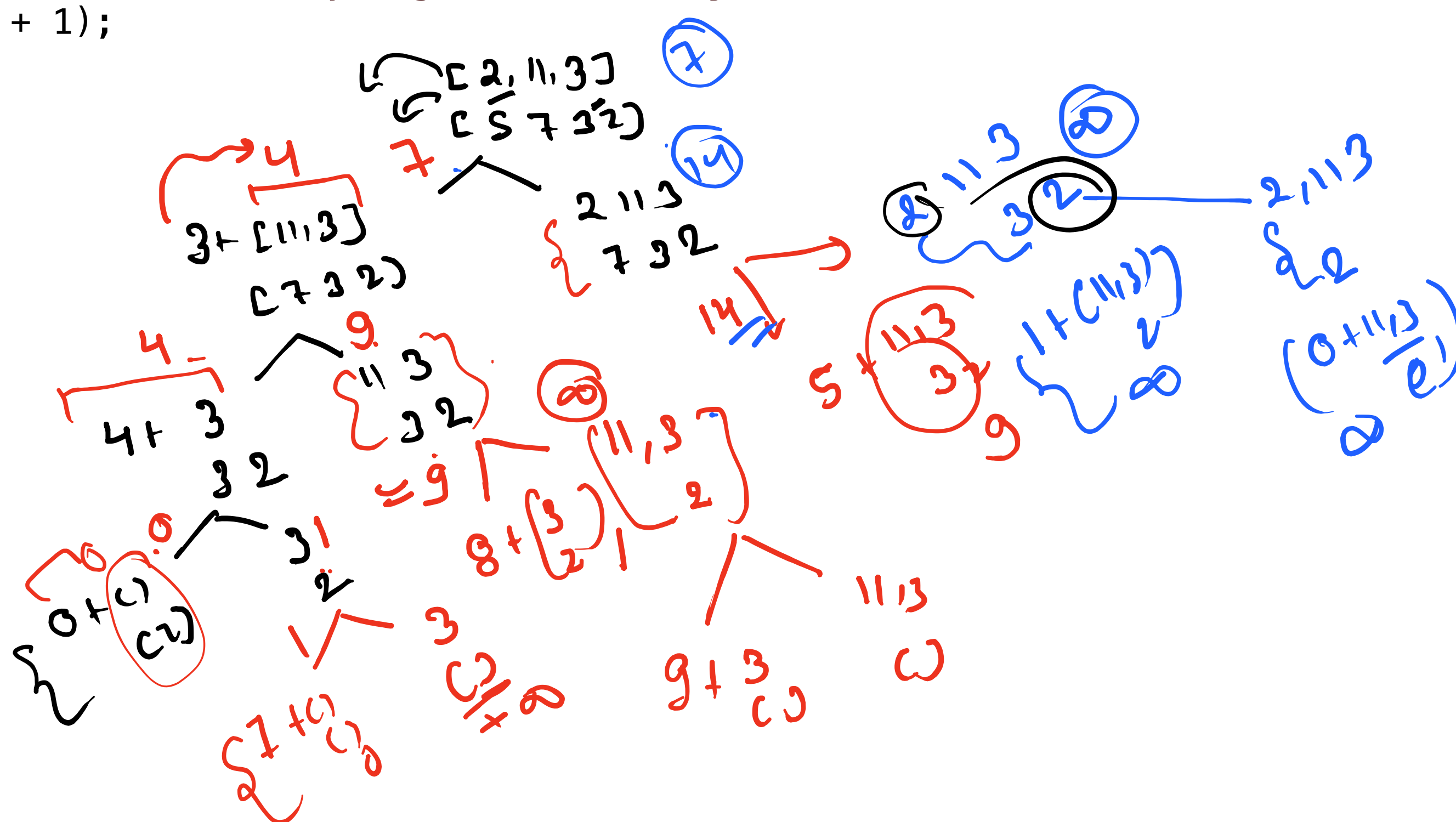
Valentine Magic

It's Valentine's Day in Russia today, as we all know number of girls in Russia is more than number of boys 🥰 hence boys have an extra advantage while choosing girl for the valentine evening. Each boy has certain number of chocolates and each girl has certain number of candies. Now you being the anchor of evening wants to pair all the boys with girls such that the sum of absolute difference between boy's chocolate and girl's candy in a pair is minimized. Ofcourse some of the girls will remain unpaired but that's okay as we are in Russia 🥰



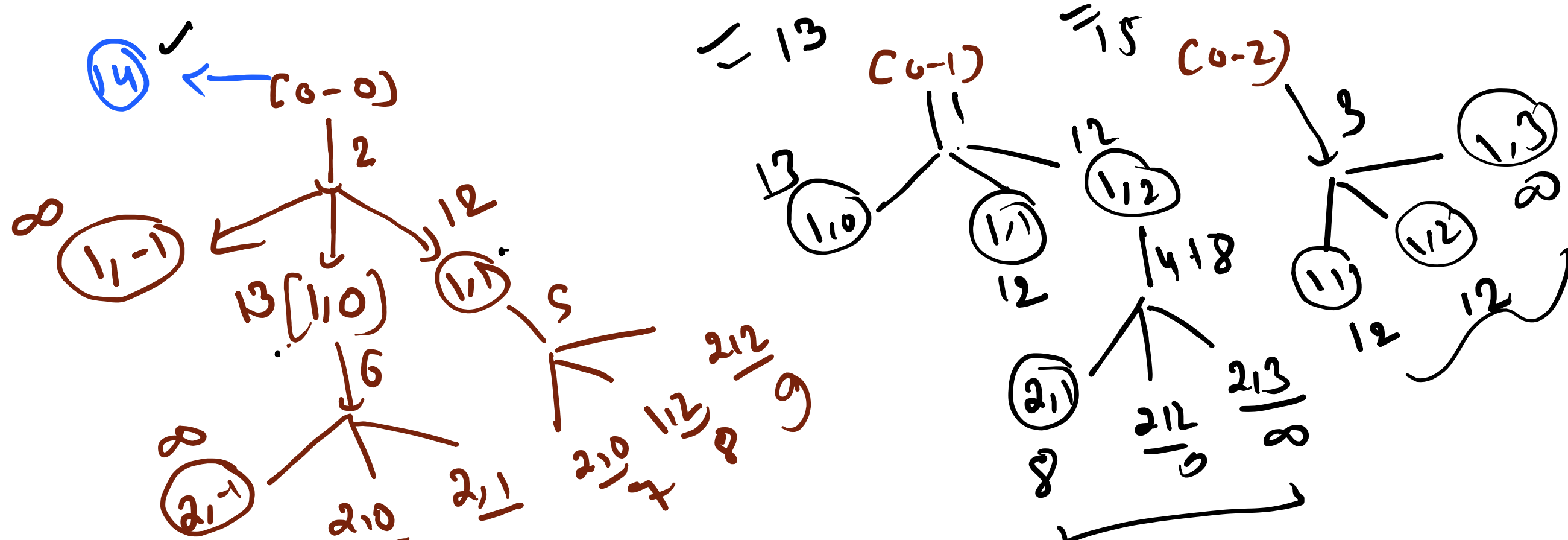
```
public static int Valentine(int[] boys, int[] girls, int i, int j) {
    if (i == boys.length) {
        return 0;
    }
    if (j == girls.length) {
        return 9999999;
    }

    int sel = Math.abs(boys[i] - girls[j]) + Valentine(boys, girls, i + 1, j + 1);
    int rej = Valentine(boys, girls, i, j + 1);
    return Math.min(sel, rej);
}
```



	0	1	2
0	2	1	3
1	6	5	4
2	7	8	9

2	1	3
6	5	4
7	8	9



1	2	3
4	5	6
7	8	9

